

Junzhao Yang

Pittsburgh, PA | +1 (412) 844-1653 | junzhaoy@andrew.cmu.edu | fizzydavid.github.io

EDUCATION

Carnegie Mellon University, Computer Science Department 2024–Present
Ph.D. in Computer Science (expected 2027). Advisor: Richard Peng
Research areas: numerical linear algebra, spectral graph theory

Tsinghua University, Institute for Interdisciplinary Information Sciences 2020–2024
B.S. in Computer Science, Yao Class.

PROJECTS

FizzyProver May 2026–Present
Sole developer · Multi-agent system for mathematical research

- Designed a recursive Codex workflow that schedules independent proof subproblems in parallel and alternates prover and verifier passes to critique and revise proof attempts.
- Built and deployed a web platform used by 20+ academic users across 300+ runs. Among the runs we have examined to date, including runs powered by GPT-5.5, we have not observed any incorrect final proofs; by contrast, our direct GPT-5.5 Pro attempts produced flawed proofs.

HONORS AND AWARDS

- ICPC North America Championship 2025, 2nd place, Silver Medal.
- 46th ICPC World Finals, 10th place, Bronze Medal** (held in 2024).
- International Olympiad in Informatics 2019, 5th place, Gold Medal.**
- National Olympiad in Informatics 2018, Gold Medal.
- Yao Award, Bronze Medal (awarded to five Yao Class students in 2023).

PUBLICATIONS

Authors are listed alphabetically, as is customary in theoretical computer science.

- | | |
|---|-----------------------|
| [1] Fast Convergence of Gradient Descent on Random ℓ_p-Norm Regression
Yang Liu, Richard Peng, Weina Wang, Junzhao Yang | FOCS 2026 (to appear) |
| [2] Approximate Spanning Tree Counting from Uncorrelated Edge Sets
Yang P. Liu, Richard Peng, Junzhao Yang | FOCS 2026 (to appear) |
| [3] Entrywise Approximate Solutions for SDDM Systems in Almost-Linear Time
Angelo Farfan, Mehrdad Ghadiri, Junzhao Yang | STOC 2026 |
| [4] Entrywise Approximation for Matrix Inversion and Linear Systems
Mehrdad Ghadiri, Hoai-An Nguyen, Junzhao Yang | SODA 2026 |
| [5] Numerical Linear Algebra in Linear Space
Yiping Liu, Hoai-An Nguyen, Junzhao Yang | SODA 2026 |
| [6] Memory-Sample Lower Bounds for LWE
Mingqi Lu, Junzhao Yang | CRYPTO 2024 |
| [7] The Cost of Parallelizing Boosting
Xin Lyu, Hongxun Wu, Junzhao Yang | SODA 2024 |
| [8] Tight Time-Space Lower Bounds for Constant-Pass Learning
Xin Lyu, Avishay Tal, Hongxun Wu, Junzhao Yang | FOCS 2023 |